

Introduction to Microeconomics

Lecture 10—Consumer Theory: The Consumer's Problem

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The Consumer's Problem

The Consumer's Problem

Choose the best bundle, according to the consumer's preferences subject to the consumer's budget constraint. → Choose the best among what is feasible.

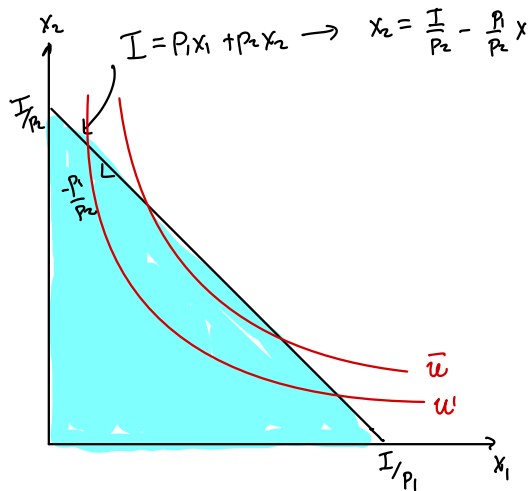
$$\begin{aligned} \max_{(x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n} & \quad u(x_1, x_2, \dots, x_n) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq I \end{aligned}$$

where $p_1, p_2, \dots, p_n$ are the prices of the n goods and I is income.

Two-good world

$$\begin{aligned} \max_{(x_1, x_2) \in \mathbb{R}_+^2} & \quad u(x_1, x_2) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 \leq I \end{aligned}$$

Graphical Representation



$$I \geq p_1 x_1 + p_2 x_2$$

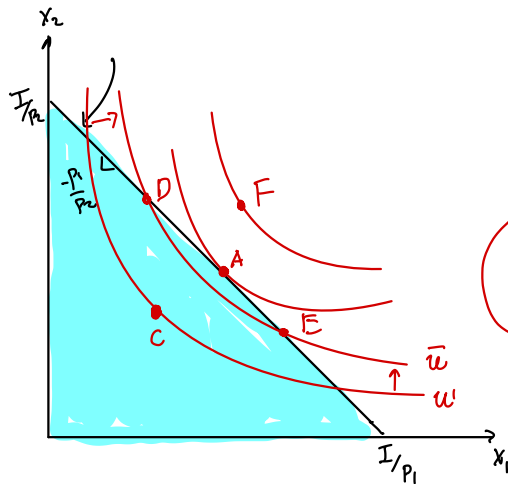
Feasible consumption bundles for the consumer.

Preferences are represented via the indifference curve map

$$IC_{\bar{u}} = \{(x_1, x_2) \mid \bar{u} = u(x_1, x_2)\}$$

$$u(x_1, x_2) = x_1 x_2 \rightarrow \bar{u} = x_1 x_2 \\ \rightarrow x_2 = \frac{\bar{u}}{x_1}$$

Graphical Representation



Point A is the optimal consumption bundle.

① $MRS = \frac{P_1}{P_2}$

② $p_1 x_1 + p_2 x_2 =$

Identify an optimal consumption bundle.

$$\frac{MU_1}{MU_2} = \frac{P_1}{P_2} \Leftrightarrow \frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$

Optimality in the Consumer's Problem

$$\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2) \quad \text{subject to: } p_1 x_1 + p_2 x_2 \leq I$$

Suppose that preferences are:

- ① Complete
- ② Transitive
- ③ Monotonic (More is better)
- ④ Strictly Convex (strict preference for balancedness) $\leftrightarrow$ (MRS strictly decreasing)

The optimality condition

We can simplify the problem to: $\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2)$ subject to: $p_1 x_1 + p_2 x_2 = I$

Furthermore, the consumer's optimal consumption bundle satisfies:

$$\boxed{MRS = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}}$$

 Identify the optimal consumption bundle

$$\max_{(x_1, x_2) \in \mathbb{R}_+^2}$$

$$u(x_1, x_2)$$

① — ④ are true.

s.t.

$$p_1 x_1 + p_2 x_2 = I$$

$$\max_{x_1 \in \mathbb{R}_+}$$

$$u\left(x_1, \frac{I}{p_2} - \frac{p_1}{p_2} x_1\right)$$

$$x_1 \in \mathbb{R}_+$$

Do this

Later !

Examples

Cobb-Douglas Utility: $u(x_1, x_2) = x_1 x_2$

$$p_1 = 1, \quad p_2 = 2, \quad I = 80$$

$$MU_1 = x_2$$

$$MU_2 = x_1$$

$$\max x_1 x_2$$

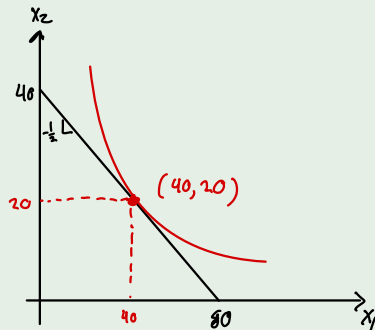
$$(x_1, x_2) \in \mathbb{R}_+^2$$

$$\text{s.t. } x_1 + 2x_2 = 80$$

$$\textcircled{1} \quad MRS = \frac{MU_1}{MU_2} = \frac{x_2}{x_1} = \frac{1}{2} \Rightarrow x_2 = \frac{1}{2} x_1$$

$$\textcircled{2} \quad x_1 + 2x_2 = 80$$

$$x_1 + 2\left(\frac{1}{2}x_1\right) = 80 \Rightarrow 2x_1 = 80 \\ x_1^* = 40$$



$$x_2^* = \frac{1}{2}(40) = 20$$

Examples

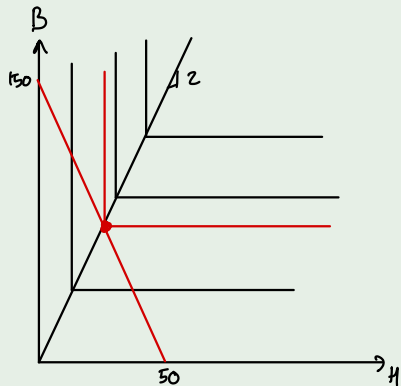
Perfect Complements *MRS is not defined.*

Suppose that I like to prepare my sandwiches with exactly 2 slices of Bread and 1 slice of Ham.

$$U(H, B) = \min \{ H, B/2 \}$$

$$\begin{aligned} &\hookrightarrow \text{Kink } H = B/2 \\ &\rightarrow 2H = B \end{aligned}$$

$$P_H = 3, \quad P_B = 1, \quad I = 150$$



$$\max_{(H, B) \in \mathbb{R}_+^2} \min \{ H, B/2 \}$$

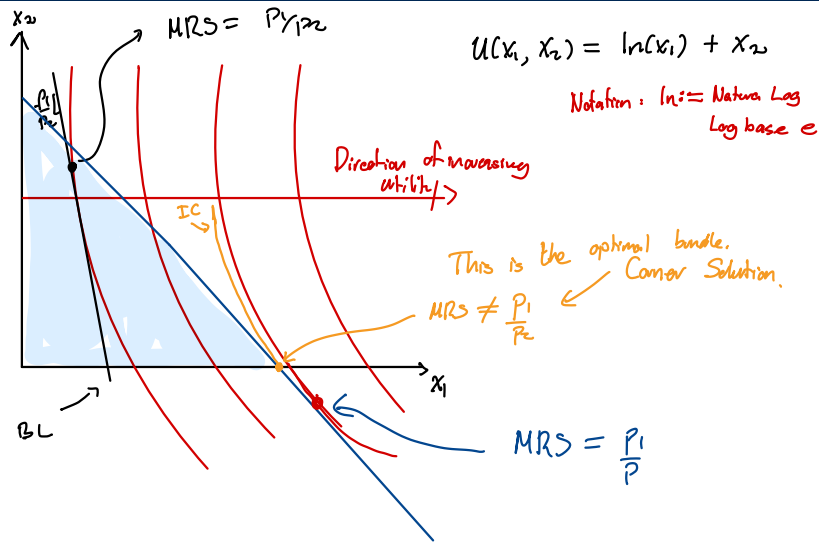
$$\begin{aligned} &\rightarrow 3H + B = 150 \\ &2H = B \end{aligned}$$

$$H^* = \frac{150}{5} = 30$$

$$3H + 2H = 150 \quad B^* = 60$$

$$5H = 150$$

Special Cases: Corner Solutions



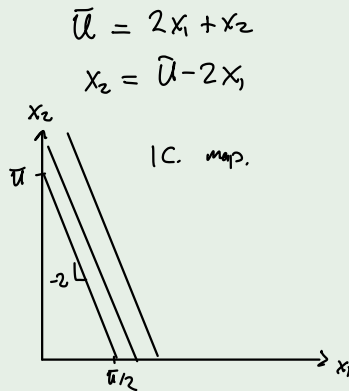
More special cases with Corner Solutions

Perfect **Substitutes** $u(x_1, x_2) = 2x_1 + x_2$

$$MRS = \frac{MU_1}{MU_2} = \frac{2}{1} = 2$$

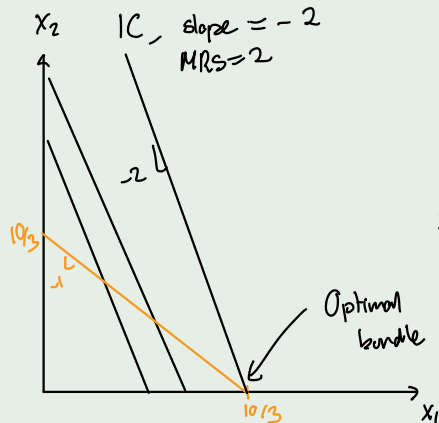
$$MU_1 = \frac{\Delta u}{\Delta x_1} = \frac{2x_1' + 2\cancel{x_2} - (2x_1 + 2\cancel{x_2})}{x_1' - x_1} \\ = \frac{2(x_1' - x_1)}{x_1' - x_1} = 2$$

$$MU_2 = \frac{\Delta u}{\Delta x_2} = 1$$



More special cases with Corner Solutions

Perfect Substitutes $u(x_1, x_2) = 2x_1 + x_2$



$$I = 500 \text{ KZT}$$

$$P_1 = 150 \text{ (price of coke)}$$

$$P_2 = 150 \text{ (price of pepsi)}$$

$$BL: 150x_1 + 150x_2 = 500$$

$$x_2 = \frac{500}{150} - x_1 = \frac{10}{3} - x_1$$

$$(10/3, 0)$$

$$x_1^* = 10/3$$

$$x_2^* = 0$$

$$MRS = 2 > 1 = \frac{P_1}{P_2}$$

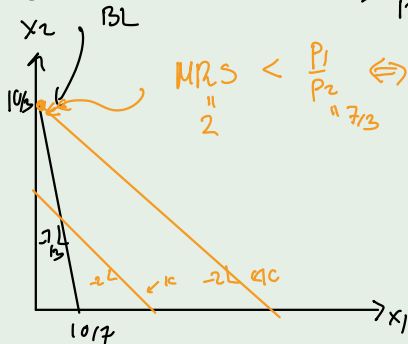
$$\hookrightarrow \frac{MU_1}{P_1} > \frac{MU_2}{P_2}$$

More special cases with Corner Solutions

Perfect **Substitutes** $u(x_1, x_2) = 2x_1 + x_2$

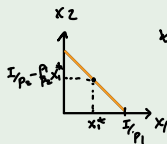
$p_1 = 350$ price of coke
 $p_2 = 150$ price of pepsi

$$350x_1 + 150x_2 = 500 ; \frac{p_1}{p_2} = \frac{350}{150} = \frac{7}{3} \approx 2.33$$



$$MRS < \frac{p_1}{p_2} \Leftrightarrow \frac{MU_1}{p_1} < \frac{MU_2}{p_2}$$

" 2 " 7/3



Prices are p_1, p_2, I

$$MRS = 2$$

$$\bullet \frac{p_1}{p_2} < 2 \quad \begin{aligned} x_1^* &= I/p_1 \\ x_2^* &= 0 \end{aligned}$$

$$\frac{p_1}{p_2} = 2$$

$$x_1^* \in [0, I/p_1]$$

$$x_2^* = \frac{I}{p_2} - \frac{p_1}{p_2} x_1^*$$

$$\bullet \frac{p_1}{p_2} > 2$$

$$x_1^* = 0$$

$$x_2^* = I/p_2$$

Example: $U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$

$$MU_1 = \frac{1}{2\sqrt{x_1}}$$

$$MRS = \frac{\sqrt{x_2}}{\sqrt{x_1}} = \frac{MU_1}{MU_2}$$

$$MU_2 = \frac{1}{2\sqrt{x_2}}$$

Optimal bundle

①

$$\frac{\sqrt{x_2}}{\sqrt{x_1}} = \frac{1}{2} \Rightarrow (\sqrt{x_2})^2 = \left(\frac{1}{2}\sqrt{x_1}\right)^2$$

$$\Rightarrow x_2 = \frac{1}{4} x_1$$

②

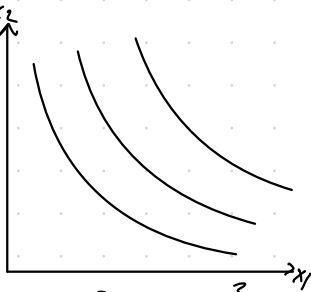
$$x_1 + 2x_2 = 100$$

$$x_1 + \frac{2}{4} x_1 = 100$$

$$\frac{6}{4} x_1 = 100$$

$$x_1^* = \frac{400}{6}$$

$$x_2^* = \frac{100}{6}$$

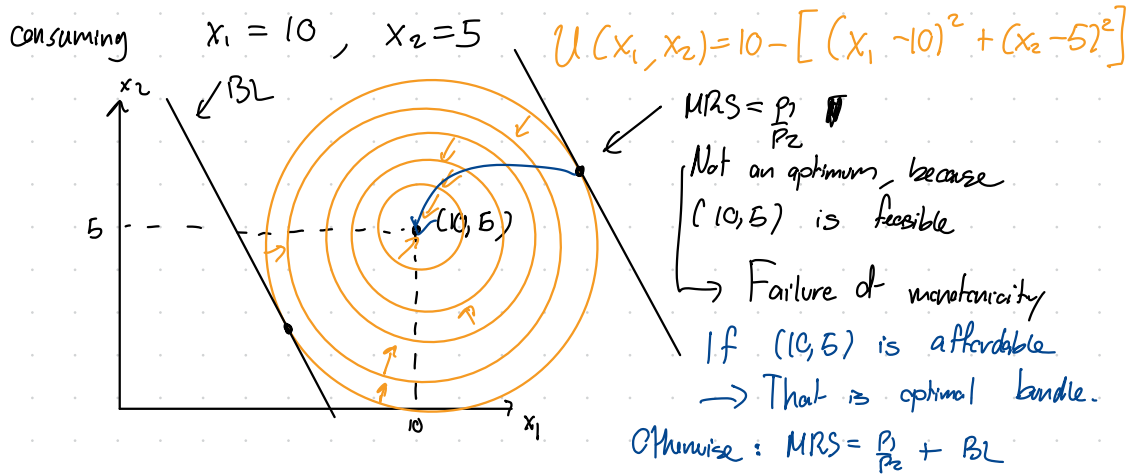


$$\bar{U} = \sqrt{x_1} + \sqrt{x_2}$$

$$x_2 = (\bar{U} - \sqrt{x_1})^2$$

Example: Consumer that is satiated (Bliss Point)

I like goods x_1 & x_2 , but I'm at my happiest when



$$\begin{array}{l}
 \max U(x_1, x_2) \\
 (x_1, x_2) \in \mathbb{R}_+^2 \\
 p_1 x_1 + p_2 x_2 = I
 \end{array}
 \left. \vphantom{\begin{array}{l} \max U(x_1, x_2) \\ (x_1, x_2) \in \mathbb{R}_+^2 \\ p_1 x_1 + p_2 x_2 = I \end{array}} \right\} \begin{array}{l} \text{Optimal} \\ \Rightarrow \\ \text{Bundle} \end{array}$$

$$\begin{array}{l}
 x_1^*(p_1, p_2, I) \\
 x_2^*(p_1, p_2, I)
 \end{array}
 \underbrace{\hspace{10em}}$$

Demand system of
the consumer.